

Problem Set on Frictional, Rationing, and Efficient Unemployment

Pascal Michaillat

Problem 1

Consider a matching model with a labor force of size 1. The matching function is Cobb-Douglas:

$$m(U, V) = \sqrt{U \cdot V},$$

where U is the number of unemployed workers and V is the number of vacant jobs. Firms have a production function

$$y(N) = 2 \cdot a \cdot \sqrt{N},$$

where $a \leq 1$ governs labor productivity and N denotes the number of producers in the firm. All workers are paid at a wage

$$w = \sqrt{a}.$$

Firms incur a recruiting cost of $r > 0$ recruiters per vacancy and face a job-destruction rate $s > 0$. The labor market tightness is $\theta = V/U$ and the employment level is $L = 1 - U$.

- A) Compute the job-finding rate $f(\theta)$ and vacancy-filling rate $q(\theta)$. Assuming that labor-market flows are balanced, compute the recruiter-producer ratio $\tau(\theta)$. Compute the elasticities of f , q , and τ with respect to θ . Interpret the signs of the elasticities.
- B) Assuming that labor-market flows are balanced, compute labor supply $L^s(\theta)$. Compute the elasticity of L^s with respect to θ . Interpret the sign of the elasticity.
- C) Firms choose employment to maximize flow profits:

$$y(N) - [1 + \tau(\theta)] \cdot w \cdot N.$$

Compute the labor demand $L^d(\theta, a)$ by solving this maximization problem. Compute the elasticities of L^d with respect to θ and with respect to a . Interpret the signs of these elasticities.

- D) Characterize tightness $\theta(a)$ and employment $L(a)$ in the model. Compute the elasticities of $\theta(a)$ and $L(a)$ with respect to a . Interpret the signs of these elasticities.
- E) Would shocks to labor productivity a create realistic business cycles?

- F) Compute the amount of rationing unemployment $U^r(a)$ and frictional unemployment $U^f(a)$ in the model.
- G) Prove that $dU^f/da > 0$. Interpret the result and provide some policy implications.

Problem 2

Consider an economy with a mass 1 of participants in the labor force. The Beveridge curve takes a very simple form:

$$v(u) = \frac{\omega}{u},$$

where $\omega > 0$ governs the location of the Beveridge curve. Each vacancy requires the attention of a full-time worker. Finally, all production takes place in firms and there is no home production at all. As a result, social welfare is determined by the number of producers in firms.

- A) Compute the socially efficient labor market tightness θ^* . How does θ^* depend on the parameter ω ?
- B) Compute the socially efficient unemployment rate u^* as a function of the actual unemployment and vacancy rates, u and v .
- C) Using the formulas derived above, compute the efficient tightness, efficient unemployment rate, and unemployment gap in the United States in December 2021. What are the policy implications of your results?